MILITARY SPECIFICATION

SWITCH, INERTIA, FUZE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Fuze Inertia Switch, referred to herein as the switch.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-45662	Calibration System Requirements.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1336

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1246	Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1695	Environments, Working, Minimum Standards for.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL 1186AS221	Switch, Inertia, Fuze.

Naval Sea Systems Command
(Code Ident 53711)

LD 533577	Impact Test Fixture for Fuze Triggering Devices.
LM 2186956	Fixture, Vibration, for Fuze Triggering Devices.
LM 2203492	Fixture, Vibration, for Fuze Triggering Devices.
DL 2205011	Test Set, Vibration.
DL 2238186	Test Set, Inertia.

(Copies of specifications, standards, publications, and drawings required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178 Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

NOTE: Drawings and associated documents (such as DLs, LMs, and LDs) referenced herein which are not listed in Section 2 are delineated on the DLs, LMs or LDs cited in Section 2.

3.1 Item description. The switch contains two vibration-dampened contacts which are designed to momentarily close at impact. The contacts are located within a housing that also serves as a portion of the switching common, or ground circuit. The switch must be specifically oriented with respect to the longitudinal centerline and direction of travel of the fuze or missile in which it is mounted. Upon impact of the missile, a circuit for delivery of a firing pulse to initiate the explosive-train of the missile warhead is completed through closure of either one or both contacts of the switch.

3.2 Characteristics.

3.2.1 Performance. The performance characteristics of the switch shall be as specified herein when fabricated in accordance with the drawings and documents listed on DL 1186AS221.

3.2.1.1 Insulation resistance. The insulation resistance between the switch housing (Drawing 1186AS222) and both leads shorted together shall be 25 megohms (Mohms) minimum.

3.2.1.2 Impact.

3.2.1.2.1 Impact, 65-degree position. Neither contact of the switch shall close when subjected to a 65-degree impact in accordance with Figure 1.

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3.2.1.2.2 Impact, 95-degree position. Both contacts of the switch shall close when subjected to a 95-degree impact in accordance with Figure 1.

3.2.1.3 Acceleration.

3.2.1.3.1 Acceleration, 350 gravity units (g). Neither contact of the switch shall close at an acceleration of 350 +5, -0 g.

3.2.1.3.2 Acceleration, 450 g. Both contacts of the switch shall close at an acceleration of 450 +0, -5 g.

3.2.2 Environmental conditions. The switch shall not be damaged nor shall performance be degraded by exposure to the following environments.

3.2.2.1 Sinusoidal vibration. Neither contact of the switch shall close during and the switch shall be operable (see 6.3.1) following exposure to the environment of 4.4.4.1.

3.2.2.2 Random vibration. Neither contact of the switch shall close during and the switch shall be operable (see 6.3.1) following exposure to the environment of 4.4.4.2.

3.2.2.3 Thermal shock. The switch shall be operable (see 6.3.1) following exposure to the environment of 4.4.4.3.

3.2.3 Visual. There shall be no evidence of external damage, such as cracks, dents, or other blemishes, which may interfere with functioning or fit of the switch at the next assembly.

3.3 First article (preproduction sample). Unless otherwise specified in the contract or purchase order, the contractor shall furnish a preproduction sample of 50 switches to the testing activity designated in the contract or purchase order for first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample shall be at the risk of the contractor.

3.4 Construction. The construction of the switch shall meet the requirements specified herein and on the drawings and documents listed on DL 1186AS221.

3.4.1 Production drawings. The switch shall meet the requirements of and be manufactured and assembled in accordance with the drawings and documents listed on DL 1186AS221.

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3.4.2 Test equipment calibration plan. When specified in the contract or purchase order, test equipment shall be calibrated and maintained in accordance with a plan meeting the requirements of MIL-C-45662. The calibration plan shall be prepared in accordance with the data item description (DID) cited in 6.2.2 and shall be approved by the Government prior to use.

3.4.3 Test plan and test procedures. When specified in the contract or purchase order, the contractor shall prepare a test plan and test procedures in accordance with the DID cited in 6.2.2. The test plan and test procedures shall be approved by the Government prior to use.

3.4.4 Inspection record When specified in the contract or purchase order, the contractor shall prepare an inspection record in accordance with the DID cited in 6.2.2 for each switch produced under a specified contract. The required test data shall be extracted from the inspection and acceptance records maintained by the contractor.

3.4.5 Standards of manufacture.

3.4.5.1 Product cleanliness. Product cleanliness shall meet the requirements of MIL-STD-1246, Level 200. No oils or solid lubricants of any kind shall be used on any part of the switch after final cleaning of components, except as specified on the drawings or herein.

3.4.5.2 Working environment. The working environment shall meet the minimum levels and requirements specified in MIL-STD-1695.

3.4.6 Workmanship. The switch shall be constructed and finished in a manner that ensures conformance to the requirements specified on the drawings and herein.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction sample inspections (see 4.1.2.1).

b. Quality conformance inspections (see 4.2).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction sample inspections. The preproduction sample of 50 switches shall be subjected to the inspections of Figure 2. The failure of any switch to meet all the requirements specified in the contract or purchase order, on the drawings, or herein shall cause rejection of the preproduction sample. The inspections below the dashed line of Figure 2 shall be performed at an independent facility designated in the contract or purchase order (see 6.2.1).

4.2 Quality conformance inspections.

4.2.1 Inspection lot. Unless otherwise specified in the contract or purchase order, an inspection lot shall contain not less than 151 nor more than 500 switches (see 6.2.1). An inspection lot, as defined in MIL-STD-105, shall consist of completed switches fabricated under the same conditions and offered for inspection at the same time

4.2.1.1 Serial numbers. Unless otherwise specified in the contract or purchase order, all switches shall be assigned a Government serial number in order of consecutive assembly (see 6.2.1). After assignment of a serial number to each switch, the inspection lot shall be formed and identified by lot number.

4.2.2 Quality conformance inspection sequence. After the inspection lot has been formed and identified by number, the lot shall be subjected to the inspection sequence above the dashed line of Figure 3. Fifty switches shall then be randomly sampled from the inspection lot in accordance with MIL-STD-105 and subjected to the inspections below the dashed line of Figure 3 at an independent facility designated in the contract or purchase order (see 6.2.1). The remainder of the lot shall be held in bond pending acceptance of the 50-switch sample. The 50-switch sample shall not be returned to the lot nor forwarded for service use, but shall be considered destroyed and disposed of at the discretion of the contracting officer.

4.2.2.1 Quality conformance acceptance and rejection criteria. Defects in the switch are classified as MAJOR, as defined in MIL-STD-105, and are specified on the drawings. Quality conformance acceptance and rejection criteria shall be as follows:

a. For the inspections above the dashed line of Figure 3 where 100 percent inspection is required, rejection shall be on an individual basis. The failure of any switch to meet the requirements specified in the contract or purchase order, on the drawings, or herein shall cause rejection of the switch.

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b. For the inspections below the dashed line of Figure 3 where random sampling is required, the inspection lot shall be rejected when the number of MAJOR defects found exceeds an acceptable quality level (AQL) of 0.25 percent defective for switch closure during the random vibration test of 4.4.4.2, or an AQL of 1.0 percent defective for the tests of 4.4.1, 4.4.2, and 4.4.3.

4.3 Inspection conditions.

4.3.1 Inspection environment. Unless otherwise specified herein, the inspections of 4.4 shall be performed under the conditions specified in 3.4.5.2.

4.3.2 Test equipment and inspection facilities. The contractor shall furnish all necessary test equipment, inspection facilities, and personnel necessary to perform the required acceptance tests. The test equipment shall be of sufficient accuracy and quality to permit satisfactory performance of the required tests and examinations, and shall be calibrated as specified in 3.4.2.

4.3.3 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. When specified in the contract or purchase order, the procuring activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.4 Methods of inspection.

4.4.1 Insulation resistance. Test the switch in accordance with MIL-STD-202, Method 302, Test Condition B. Connect the megohmmeter leads between the shorted lead wires of the switch and housing. Record the resistance. The insulation resistance shall conform to 3.2.1.1.

4.4.2 Impact. For the tests of 4.4.2.1 and 4.4.2.2, mount the switch on the impact test fixture (LD 533577) at the base of the mass in the position shown on Figure 1, so as to receive the impact in the forward direction (see Drawing 1186AS221) along the longitudinal axis (see 6.3.2). Use the eight-channel inertia test set (DL 2238186) to individually monitor closure of both contacts.

NOTE: Contact closure will be indicated by the illumination of test lamps on the inertia test set (Drawing 2238186).

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4.4.2.1 Impact, 65-degree position. Raise the pendulum to the 65-degree position, as measured by the indicator wheel shown on Figure 1. Release the pendulum and observe the monitor lamps on the inertia test set to determine conformance to 3.2.1.2.1.

4.4.2.2 Impact, 95-degree position Raise the pendulum to the 95-degree position, as measured by the indicator wheel shown on Figure 1. Release the pendulum and observe the monitor lamps on the inertia test set to determine conformance to 3.2.1.2.2.

4.4.3 Acceleration. For the tests of 4.4.3.1 and 4.4.3.2, mount the switch on a centrifuge so that acceleration will be applied in the forward direction (see Drawing 1186AS221) along the longitudinal axis (see 6.3.2).

4.4.3.1 Acceleration, 350 g. Accelerate the switch to 350 +5, -0 g, apply power to the monitoring circuits of the inertia test set, and observe the monitor lamps to determine conformance to 3.2.1.3.1. Remove power to the monitoring circuits.

4.4.3.2 Acceleration, 450 g. Accelerate the switch to 450 +0, -5 g, apply power to the monitoring circuits of the inertia test set, and observe the monitor lamps to determine conformance to 3.2.1.3.2. Remove power to the monitoring circuits.

4.4.4 Environmental.

4.4.4.1 Sinusoidal vibration. To determine conformance to 3.2.2.1, test the switch as follows:

a. Mount the vibration fixture (LM 2186956) on a vibration machine and tighten the mounting bolts to a torque of 25 ±2 foot-pounds (ft-lb).

b. Mount the switch on the fixture and tighten the mounting clamp screw to a torque of 140 ±3 inch-pounds (in-lb).

c. Connect the vibration test set (DL 2205011) to the switch to monitor contact closure (both contacts of the switch).

d. Along the longitudinal axis of the switch (see 6.3.2), apply sinusoidal vibration over a frequency range from 10 to 3,000 to 10 hertz (Hz) at an acceleration of 25 g ±10 percent, or a double-amplitude displacement of 0.50 ±0.05 inch, whichever is less severe as established prior to the test. The frequency range shall be swept at a logarithmic rate, and one complete sweep from 10 to 3,000 to 10 Hz shall take 5 ±1 minutes. Repeat the test for a total time of 10 ±2 minutes. Monitor both contacts of the switch for functioning during the test.

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4.4.4.2 Random vibration. To determine conformance to 3.2.2.2, test the switch as follows:

a. Mount the vibration fixture (LM 2203492) on a vibration machine, and tighten the mounting bolts to a torque of 25 ±2 ft-lb.

b. Mount the switch on the vibration fixture and tighten the mounting clamp screw to a torque of 140 ±3 in-lb.

c. Connect the vibration test set (DL 2205011) to the switch to monitor contact closure (both contacts of the switch).

d. Along the longitudinal axis of the switch (see 6.3.2), apply random vibration as specified in MIL-STD-810, Method 514, Procedure V, Part 2, Curve AG, for 180 ±5 minutes. Monitor both contacts of the switch for functioning during the test.

e. Perform the insulation resistance test of 4.4.1.

f. Repeat step d above, except the switch shall be vibrated along an axis normal to the longitudinal axis. Monitor both contacts of the switch for functioning during the test.

g. Repeat step e above.

4.4.4.3 Thermal shock. Test the switch in accordance with MIL-STD-202, Method 107, Test Condition A. The switch shall conform to 3.2.2.3.

4.5 Visual inspection. Visually inspect the switch to determine conformance to 3.2.3 and the general requirements of Section 3.

4.6 Drawing inspections. To determine conformance to 3.4.1, perform drawing inspections on each lot of parts, subassemblies, and assemblies of the switch during manufacture, and on the completely assembled switch in accordance with the drawings listed on DL 1186AS221. The sampling plans and acceptance criteria shall be as specified on the drawings.

4.7 Packaging inspection. Prior to shipment, inspect the packaging, packing, and marking of the switch to ensure conformance to the requirements of Section 5.

5. PACKAGING.

5.1 Preservation - packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C.

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5.1.1 Level C. Switches shall be packaged in commercially-available materials in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C.

5.2.1 Level C. Switches, packaged in accordance with 5.1.1, shall be packed in commercially-available materials in a manner that will provide protection from damage during direct shipment from the supply source to the first receiving activity. Packing and the type of containers used shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The switch is intended for use in the contact fuzing system of a guided missile.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Number, title, and date of this specification.

b. First article. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 50 switches, as specified in 3.3 and 4.1.2.1. The contracting officer should include specific instructions in all instruments regarding arrangements for examinations, tests, and approval of the first article.

c. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

d. Independent facility to perform inspections below the dashed line of Figure 2 (see 4.1.2.1).

e. Inspection lot size, if other than as specified in 4.2.1.

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f. Assignment of Government serial numbers (see 4.2.1.1).

g. Independent facility to perform inspections below the dashed line of Figure 3 (see 4.2.2).

h. Government final inspection gage drawings (see 4.3.3).

i. Special marking (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-1049(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited as follows:

Paragraph	Data requirement	Applicable DID
3.4.2	Calibration Plan	UDI-T-23742A
3.4.3	Test Plan and Test Procedures	DI-T-5204
3.4.4	Inspection Record	DI-T-2072

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Operable. Operable is defined as the ability of the switch to meet the requirements of 3.2.1.1 and 3.2.1.2.

6.3.2 Longitudinal axis. The longitudinal axis is defined as along the forward direction of the switch as shown on Drawing 1186AS221.

Preparing activity:
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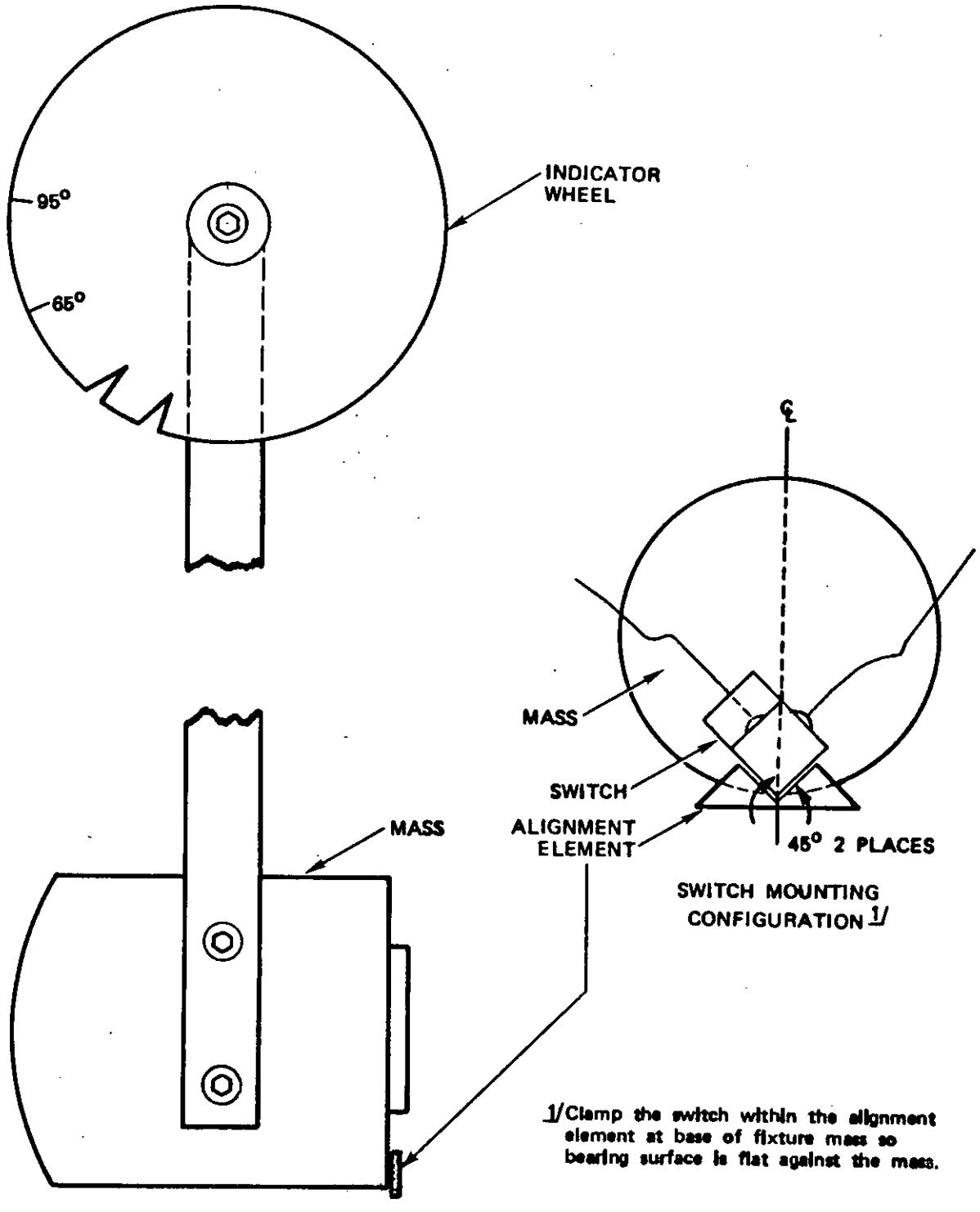


FIGURE 1. Impact test setup.

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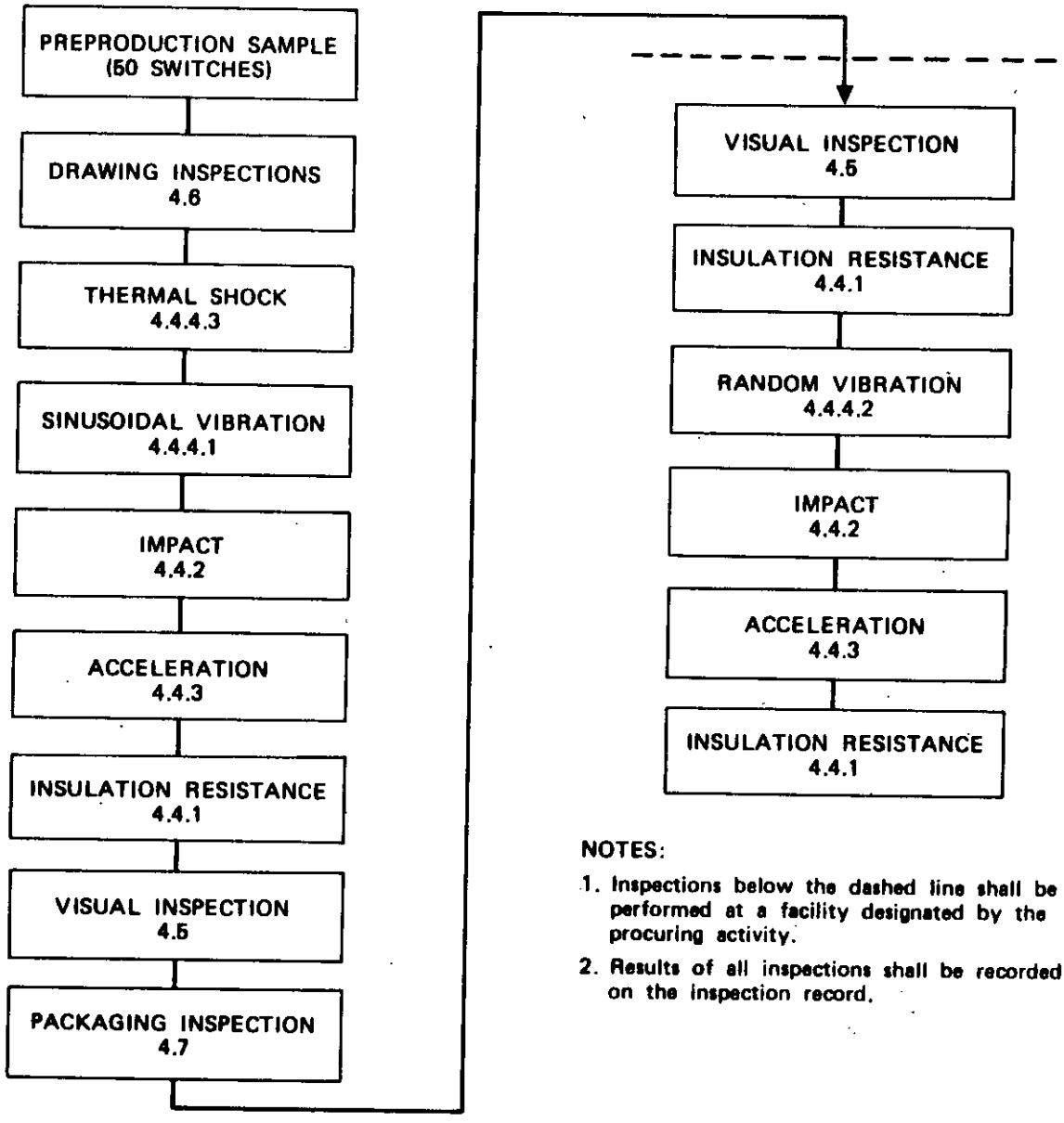


FIGURE 2. Preproduction sample inspection sequence.

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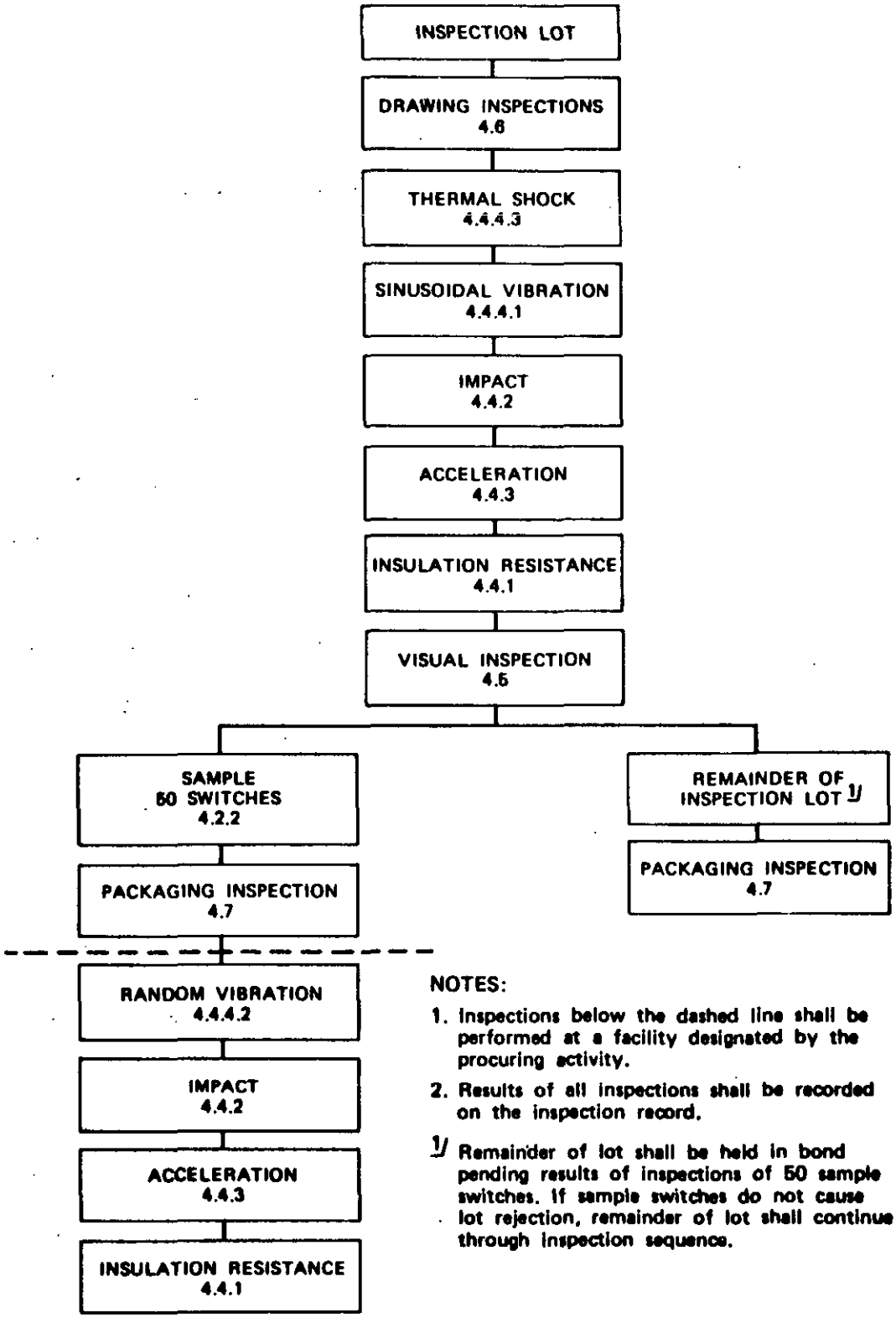


FIGURE 3. Inspection lot inspection sequence.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-S-85354(AS), Switch, Inertia, Fuze

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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